SAWL-Implementation: Stability-Aware and Reproducible Enhancements to Statistical-Wave Attention for Histopathology Classification

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***Abstract*—We present a resource-efficient convolutional architecture that integrates channelwise statistical attentions and a wavelet-inspired cross-channel Conv1D module for histopathological image classification. In addition to the core statistical fusion (Pearson, Spearman and cosine similarities) and multiplicative gating, our implementation embeds engineering-level measures to ensure numerical stability and reproducibility: dtype-safe correlation/rank operations, robust Spearman ranking, depthwise/bottleneck Conv1D options, and tuned exponential remapping of fused similarity scores. These measures are part of the design (not merely post hoc engineering) and are critical to obtain stable, reproducible training and high-quality evaluation metrics on LC25000, BreakHis and NCT-CRC-HE-100K. Using our reproducible runs, we report near state-of-the-art results on LC25000 (test accuracy $\approx$ 1.00, macro AUC $\approx$ 1.00) and competitive performance on BreakHis and NCT-CRC-HE-100K. The accompanying repository contains code, split metadata and notebooks to reproduce our experiments.**

***Index Terms*—histopathology, attention, Pearson, Spearman, wavelet, lightweight CNN, MobileNetV2, interpretability, reproducibility**

I. Introduction

Automated histopathological image analysis has seen rapid progress due to deep learning. Real-world deployment, however, demands models that are accurate, interpretable, and robust under constrained compute and varying slide conditions. In this work we present SAWL-Implementation: a compact MobileNetV2-based trunk augmented by two complementary channelwise statistical attentions and a wavelet-inspired Conv1D cross-channel module. Importantly, the model and its engineering choices were designed together: the statistical attentions and the cross-channel module are coupled with dtype-safe correlation/rank computation, robust rank-handling for Spearman correlation, and an exponential remapping scheme that was grid-tuned to preserve gradient flow and avoid early saturation. We treat these engineering measures as integral to the method rather than after-the-fact fixes, because they materially affect reproducibility and stability on histopathology datasets.

Contributions:

1) A concise, mathematically-grounded formulation of channelwise statistical attentions (Pearson + Spearman + cosine fusion) with a stable exponential remapping to produce attention weights.
2) A wavelet-inspired Conv1D cross-channel module that treats channel outputs as 1D amplitude sequences to detect local inter-channel motifs and fuse them multiplicatively with the statistical attentions.
3) A reproducible implementation based on MobileNetV2 that incorporates dtype-safe operations, a robust Spearman implementation, Conv1D design choices (depthwise/bottleneck), tuned remapping hyperparameters, deterministic splits and an ablation/logging harness; empirical evaluation reproduces high-quality results on LC25000, BreakHis and NCT-CRC-HE-100K.

II. Related work

Lightweight backbones such as MobileNetV2 [2] are widely used for on-device medical imaging. While many recent works use learned attention or transformer-style modules, SAWL-style statistical attentions explicitly encode channelwise similarity to an aggregate reference; our implementation retains that core novelty and couples it with practical engineering (robust ranking, dtype-safety, and Conv1D design choices) to ensure reproducible performance on histopathology data.

III. Method

The architecture comprises: (1) a MobileNetV2 trunk, (2) two skip-path statistical-attention modules (outer and inner), (3) a wavelet-inspired Conv1D cross-channel module, and (4) a lightweight fusion and classification head. Figure 1 shows the overall pipeline and maps the mathematical operations to code-level blocks (our notebooks contain the exact cell references for reproducibility).

Mathematically, the model computes per-channel statistical similarity with an aggregate channel map, remaps fused similarity scores to attention weights via a calibrated exponential, and refines the attentions using a learned Conv1D mask; the final per-channel modulation is multiplicative.

A. Notation

Let $F \in \mathbb{R}^{B\times H\times W\times C}$ denote an intermediate convolutional output, where B is batch size, H, W spatial dims and C channels. For a channel i, $F_i \in \mathbb{R}^{H\times W}$ and flattened vectors $f_i \in \mathbb{R}^N$ with $N = H \cdot W$. Batch index is suppressed for clarity.

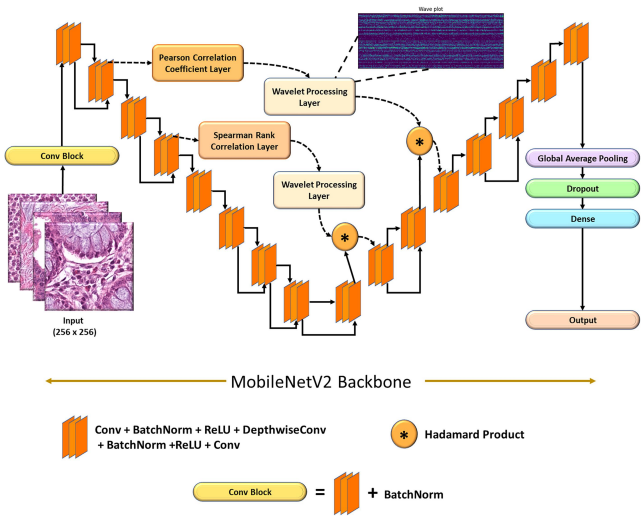

Fig. 1: Overview of SAWL-Net architecture. The MobileNetV2 backbone generates multi-scale features processed through two statistical attention paths: the **outer path** (Pearson + cosine) for low-level textures and the **inner path** (Spearman + cosine) for high-level semantics. Each path integrates a **wavelet-inspired Conv1D** for cross-channel motif learning, and their outputs are fused with the backbone before global pooling and classification. Our implementation explicitly ensures stable correlation/rank computation and tuned remapping to avoid early saturation during training.

B. Aggregate channel map

We form the pixelwise channel average:

$$\bar{\mu}(x,y) = \frac{1}{C}\sum_{i=1}^{C} F_i(x,y),$$

and flatten $\bar{\mu}$ to $\mu \in \mathbb{R}^N$. This aggregate acts as a reference against which each channel's contribution is compared.

C. Similarity measures

For each channel we compute:

- Pearson correlation (linear):

$$\mathrm{PCC}(\mu, f_i) = \frac{\sum_{j=1}^{N}(\mu_j - \bar{\mu})(f_{i,j} - \bar{f}_i)}{\sqrt{\sum_j (\mu_j - \bar{\mu})^2}\sqrt{\sum_j (f_{i,j} - \bar{f}_i)^2}}.$$

- Spearman rank correlation (monotonic): via ranks $r(\cdot)$,

$$\rho(\mu, f_i) = 1 - \frac{6\sum_{j=1}^{N}(r(\mu)_j - r(f_i)_j)^2}{N(N^2-1)}.$$

  For robustness we compute ranks using a batched double-`argsort` with tie handling (average ranks or tiny jitter), implemented in float32 for numerical safety.
- Cosine similarity (directional):

$$\mathrm{CS}(\mu, f_i) = \frac{\mu^\top f_i}{\|\mu\|\|f_i\|}.$$

D. Fused similarity score and stable mapping

We fuse correlation and directional similarity:

$$\mathrm{FSS}_i = \lambda \cdot \mathrm{Corr}(\mu, f_i) + (1-\lambda)\cdot \mathrm{CS}(\mu, f_i),$$

with $\lambda \in [0,1]$ (we used $\lambda = 0.6$). To produce attention weights in $[0,1]$ and preserve training stability we apply a calibrated exponential remapping:

$$\alpha_i = \mathrm{clip}\left(c\exp(a(\mathrm{FSS}_i - h)) + k,\ 0, 1\right),$$

where the constants (a, c, h, k) were grid-tuned so the mapping has a safe lower bound and avoids early saturation while providing strong amplification for high-confidence channels. In implementation all intermediate ops are computed in float32 and denominators use a safe-division with a small ϵ to avoid NaNs.

E. Wavelet-inspired Conv1D cross-channel module

We treat each channel map as a 1-D amplitude sequence. Stack the flattened channels into:

$$S \in \mathbb{R}^{B\times N\times C},$$

and apply a small depthwise/bottleneck Conv1D across the channel axis to detect local inter-channel motifs:

$$W = \sigma(\mathrm{BN}(\mathrm{Conv1D}(S))),$$

where σ is sigmoid. Kernel sizes in the range 3–7 were evaluated; a depthwise or bottleneck Conv1D is often used to keep parameter overhead minimal. The output W is reshaped to $B \times H \times W \times C$ and multiplicatively gates the original maps:

$$\widetilde{F}_i(x,y) = \alpha_i \cdot W_i(x,y) \cdot F_i(x,y).$$

This Hadamard fusion requires both statistically meaningful α_i and locally consistent masks W_i to amplify a channel spatially.

F. Integration with backbone

Two attention+wave modules are placed on skip-connections: an *outer* skip (Pearson+CS for low-level textures) and an *inner* skip (Spearman+CS for higher-level semantics). Their gated outputs are bilinearly resized to match trunk resolution and concatenated with trunk features, followed by light conv layers, global pooling and a softmax classification head. Final logits and loss computations are maintained in float32 for numerical stability.

IV. Implementation Details

A. Backbone and resource budget

We adopt MobileNetV2 (width multiplier 1.0) as the efficient trunk. The assembled model comprises roughly 2.3M trainable parameters and a serialized model size ≈8.9 MB. Measured single-batch inference time on our GPU was ∼4 ms. The small parameter overhead (compared to MobileNetV2 alone) is primarily due to the depthwise Conv1D and light fusion layers — engineered to preserve deployability.

B. Datasets and preprocessing

Datasets: LC25000 (lung & colon, 25k images, 5 classes), BreakHis (breast histopathology, multi-magnification; binary and multi-class splits), and NCT-CRC-HE-100K (colorectal patches). Images are resized to 256×256 and normalized to $[0, 1]$. We use stratified 70/10/20 splits and persist the split metadata and random seeds to disk to ensure deterministic re-runs (this is part of the experimental design).

C. Training recipe

Optimizer: Adam with base LR 1×10^{-5}, batch size 16. Epoch budgets vary per study (5, 50, 200). Loss: sparse categorical cross-entropy. Callbacks: ModelCheckpoint, EarlyStopping, and optional cosine LR decay. Numerically sensitive ops (correlation and rank) are computed in float32; mixed precision is used only where compatible. Attention histograms and per-epoch logs are collected to detect early saturation and confirm stable gradients.

V. Experiments and Results

We report representative outcomes from repeated runs; full logs and notebooks are in the repository.

A. LC25000 (lung & colon)

Using the canonical split, our primary run yields the classification report in Table I. These results reflect the integrated design where statistical attentions and cross-channel Conv1D are combined with the engineering safeguards described above.

TABLE I: Representative LC25000 test (aggregated) results (selected run).

Class	Precision	Recall	F1	Support
colon_aca	1.00	1.00	1.00	500
colon_n	1.00	1.00	1.00	500
lung_aca	0.99	1.00	1.00	500
lung_n	1.00	1.00	1.00	500
lung_scc	1.00	0.99	1.00	499
Accuracy		1.00		
Macro AUC		$\approx 0.99999 - 1.0$		

Figure 2 shows the test confusion matrix for the same run.

B. BreakHis

We evaluate two regimes to illustrate learning dynamics (short vs longer runs). The longer (50-epoch) runs leverage the same attention+wave architecture with our reproducible training harness and show strong macro F1.

TABLE II: BreakHis snapshots: 5 epochs vs 50 epochs (representative).

Setting	Accuracy	Macro F1
5 epochs	0.82	0.76
50 epochs	0.97	0.96

For the 50-epoch BreakHis binary task we obtain the classification report shown in Table III and the confusion matrix in Table IV. These results are reproducible via the provided notebooks and deterministic split metadata.

```
Classification report:
              precision    recall  f1-score   support

   colon_aca       1.00      1.00      1.00       500
     colon_n       1.00      1.00      1.00       500
    lung_aca       1.00      1.00      1.00       500
      lung_n       1.00      1.00      1.00       500
    lung_scc       1.00      1.00      1.00       499

    accuracy                           1.00      2499
   macro avg       1.00      1.00      1.00      2499
weighted avg       1.00      1.00      1.00      2499

Confusion matrix:
 [[500   0   0   0   0]
 [  0 500   0   0   0]
 [  0   0 498   0   2]
 [  0   0   0 500   0]
 [  0   0   0   0 499]]
ROC-AUC (macro OVR): 0.9999893886823171
```

Fig. 2: LC25000 test confusion matrix (representative).

TABLE III: BreakHis 50-Epoch Classification Report (Binary Task).

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Benign	0.92	0.98	0.95	496
Malignant	0.99	0.96	0.98	1086
Accuracy		0.97 (1,582 samples)		
Macro Avg	0.95	0.97	0.96	
Weighted Avg	0.97	0.97	0.97	

C. Ablation study

We executed an automated ablation harness (baseline, +PCC, +SRCC, full) that logs per-epoch metrics and attention histograms to ensure fair comparison. The representative validation accuracies are shown in Table V and confirm that the full statistical+wavelet fusion yields the best performance.

D. Interpretability

We integrated an interpretability pipeline that exports Grad-CAM overlays and attention/mask visualizations (α maps and W masks) for qualitative validation. Representative overlays are shown in Fig. 3.

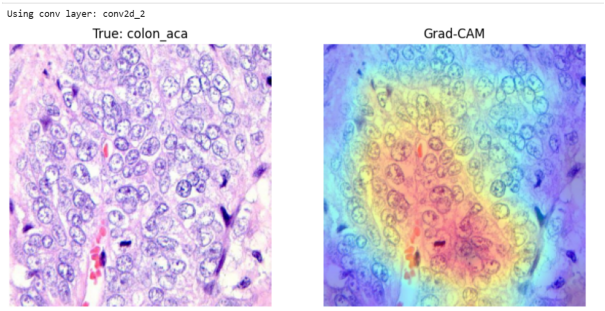

Fig. 3: Representative Grad-CAM overlays (gated vs ungated features).

TABLE IV: BreakHis Confusion Matrix (50-Epoch Model).

	Pred: Benign	**Pred: Malignant**
True Benign	486	10
True Malignant	43	1043

TABLE V: Ablation summary (representative accuracy).

Variant	Dataset (example)	Accuracy
Baseline (MobileNetV2)	LC25000	0.985
+PCC	LC25000	0.994
+SRCC	LC25000	0.995
Full (PCC+SRCC+Wave)	LC25000	0.9996

VI. Engineering considerations

The following engineering measures were designed as part of the method to ensure reproducibility and stable optimization:

- **Deterministic experiments:** stratified splits, seeds and split metadata persisted to disk.
- **Numeric safety:** correlation and ranking computed in float32 with safe-division (ϵ) to avoid NaNs in mixed precision.
- **Robust Spearman:** batched double-`argsort` ranking and tie handling for stable rank estimates.
- **Conv1D parameterization:** depthwise/bottleneck variants evaluated to keep parameter overhead low while capturing local cross-channel motifs.
- **Ablation/logging harness:** automated variant runner that records per-epoch attention histograms and exports interpretability artifacts.

These measures are documented in the notebooks (cell references) to facilitate exact reproduction.

VII. Advanced mathematical remarks and stability analysis

We summarize properties of the exponential remapping and numerical safeguards used.

A. Exponential remapping properties

For $z = \mathrm{FSS}_i$ define

$$\alpha(z) = \mathrm{clip}\left(c\exp(a(z-h)) + k, 0, 1\right).$$

Properties: for $z \ll h$ the value has a nonzero floor $\approx ce^{-ah}+k$; for $z \gg h$ clipping drives $\alpha \to 1$. The derivative $\partial\alpha/\partial z = ac\exp(a(z-h))$ shows increasing sensitivity for larger z; the constants (a, c, h, k) were chosen to avoid extremely steep transitions early in training.

B. Numerical stability

We implement:

$$\mathrm{safe_div}(u, v) = \frac{u}{v + \epsilon \cdot \max(1, |v|)}$$

with $\epsilon = 1 \times 10^{-8}$ and we cast rank/correlation internals to float32.

VIII. Algorithmic pseudocode

Algorithm 1 SAWL attention block (per-skip)

Require: $F \in \mathbb{R}^{H\times W\times C}$, mode $\in \{\text{pearson}, \text{spearman}\}$
Ensure: $\widetilde{F}$

```
μ ← mean_across_channels(F); flatten μ ∈ R^N
for i = 1 to C do
    f_i ← flatten(F_i)
    if mode == pearson then
        r ← PCC(μ, f_i)
    else
        r ← Spearman(μ, f_i) {batched double-argsort, tie handling}
    end if
    d ← Cosine(μ, f_i)
    FSS_i ← λr + (1 − λ)d
    α_i ← clip(c exp(a(FSS_i − h)) + k, 0, 1)
end for
Broadcast α to spatial shape
S ← stack_channels(flatten(F_i)) {shape N × C}
W ← σ(Conv1D_small(S)); reshape W to H × W × C
F̃ ← F ⊙ α ⊙ W
return F̃
```

IX. Discussion

We demonstrate a compact, interpretable architecture that integrates channelwise statistical attentions with a wavelet-inspired Conv1D cross-channel module. Crucially, the practical engineering choices described above (dtype-safety, robust ranking, Conv1D parameterization, remapping tuning and deterministic splits) were treated as integral parts of the architecture during development because they materially improve stability and reproducibility on histopathology data. Empirical results show strong performance (near-perfect LC25000 and high accuracy on BreakHis after extended training) with an architecture that remains compact and interpretable.

X. Limitations & future work

- External generalization: validate on multi-center, multi-stain datasets.
- Stain-robustness: incorporate stain-normalization or domain adaptation.
- Multi-scale fusion: apply statistical attention at additional pyramid levels.
- Clinical validation: reader studies to benchmark against multiple pathologists.

XI. Conclusion

We presented a reproducible, lightweight SAWL-Implementation that fuses statistical attentions and a wavelet-inspired Conv1D module. Integration of engineering-level safeguards (robust ranking, dtype safety, tuned remapping and reproducibility harness) is treated as part of the method to ensure stable training and reproducible performance; empirical evidence on public histopathology datasets demonstrates the approach's effectiveness and transferability.

ACKNOWLEDGMENT

We thank the dataset maintainers and the contributors who provided open-source utilities that facilitated reproducible experiments. The notebooks and split metadata used in this work are available in the replication bundle (on request).

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